

Stephen Huang

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EDUCATION

University of California, San Diego

GPA: 3.85/4.00

M.S. in Data Science & Machine Learning

September 2026 - June 2028

B.S. in Electrical & Computer Engineering

September 2024 - June 2026

University of California, Irvine

September 2023 - June 2024

TECHNICAL SKILLS

Languages: Python, HTML/CSS, Java, JavaScript/TypeScript, Verilog

Tools: Git, GitHub Actions, Firebase, Google Cloud Functions, Vue, Node.js, Express.js, React, Docker

Libraries: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, Matplotlib, PyBullet

EXPERIENCE

UCI Engineering Student Council | *JavaScript, Express.js, Firebase, GitHub, Docker*

Irvine, CA

Backend Software Engineer

April 2024 - September 2024

- Designed a **Firebase (NoSQL)** data model and implemented serverless **REST APIs** (Node.js and Express.js on Google Cloud Functions) to create, update, and cancel bookings
- Integrated **Google OAuth** with backend **ID token verification** and role-based route protection to prevent unauthorized reservations

UCSD Existential Robotics Laboratory | *Python, Markdown, Bash*

San Diego, CA

Machine Learning Research Intern

October 2024 - Present

- Created a **Python** robotics simulation with **PyBullet** and **PyQt** for occupancy grid mapping and A* path planning as an interactive learning module for **50+ students**
- Developed **Bash** scripts to automate **Conda** environment setup, enabling new developers and non-programmers to run and modify the robotics simulation smoothly
- Standardized tutorial website contributions through a **Markdown** style guide and reusable **HTML templates**, improving content consistency and reducing authoring overhead

PROJECTS

Google Restaurant Ratings Prediction | *Pandas, PyTorch, NumPy, Scikit-learn*

- Designed a multimodal **Meta-Relation Network** for cold-start recommendations, achieving up to **58% lower MSE** compared to other baselines on a naturally skewed dataset
- Engineered an image **data pipeline** to preprocess, cache, and reuse transformed photos, accelerating multimodal and CNN model training by **~6x**

Visual-Inertial SLAM | *OpenCV, NumPy, Scikit-learn, Python*

- Implemented **VI-SLAM** with **EKF** prediction and update on SE(3) for pose and landmark estimation from IMU measurements and stereo feature observations
- Extracted and tracked visual features from stereo image sequences using **Shi-Tomasi corner detection** and **Lucas-Kanade optical flow** for landmark association across frames

Space Typing Game | *TypeScript, Jest, Vite, GitHub*

- Architected a **TypeScript** stage-progression pipeline that loads levels from JSON via a manifest, enabling stage additions and tuning without code changes
- Built a CI/CD pipeline in **GitHub Actions** with automated **Jest** testing, build validation on PRs, and automated deployment to GitHub Pages on PR merges
- Managed **sprint workflow** by scoping milestones and maintaining the **GitHub issue backlog**, keeping deliverables on schedule, and reducing coordination overhead

RELEVANT COURSEWORK

Design & Analysis of Algorithms, Advanced Data Structures, Computer Architecture, Machine Learning, Probability & Statistics, Sensing & Estimation in Robotics, Software Engineering